

BLOCK-003 Derivation Attempt v4:

Can EDC-Internal Principles Fix the Constant C
in $\kappa_5^2 = C \cdot \sigma^{-3/4}$?

EDC Working Document

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Abstract

Derivation v3 established that dimensional analysis uniquely determines $\kappa_5^2 = C \cdot \sigma^{-3/4}$ from brane tension σ alone, but leaves the dimensionless constant C unfixed. This document tests three EDC-internal approaches to fix C without fitting to the observed Newton constant G_N^{obs} : (A) pressure-balance closure using $\sigma = 2\pi R_\xi^2 \rho_P$, (B) induced gravity with EDC UV cutoff, and (C) junction curvature normalization. **Outcome:** All three attempts fail to uniquely fix C . We prove a **No-Go Lemma**: σ , ρ_P , and R_ξ together provide only two independent scales, insufficient to determine the dimensionless ratio $C = \kappa_5^2 \sigma^{3/4}$ without an additional normalization principle.

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1 Setup and Goal

1.1 Problem Statement

From derivation v3, dimensional analysis yields the unique consistent form:

$$\kappa_5^2 = C \cdot \sigma^{-3/4} \quad (1)$$

where $[\kappa_5^2] = L^3$ and $[\sigma] = L^{-4}$ in natural units ($\hbar = c = 1$).

Goal: Determine whether EDC-internal quantities can fix the dimensionless constant C without using G_N^{obs} as input.

1.2 Available EDC Inputs

From EDC Part I and program documents, we have:

1. **Brane tension** σ with $[\sigma] = ML^{-3}T^{-2} = L^{-4}$ [BL]
2. **Plenum energy density** ρ_P with $[\rho_P] = L^{-4}$ [P]
3. **Weak length scale** $R_\xi = \hbar c/M_Z$ with $[R_\xi] = L$ [BL]
4. **Pressure-balance relation** [P]:

$$\sigma = 2\pi R_\xi^2 \rho_P \quad (2)$$

assuming membrane thickness $\delta \sim R_\xi$.

1.3 Anti-Circularity Ledger

Quantity	Proposed Source	Status	Used to Fit?
σ	Brane tension (5D action input)	[BL]	NO
ρ_P	Plenum energy density	[P]	NO
R_ξ	$\hbar c/M_Z$ (weak scale)	[BL]	NO
κ_5^2	To be derived	[OPEN]	NO
C	To be derived	[OPEN]	NO
G_N^{obs}	Cavendish experiment	[BL]	FORBIDDEN

Table 1: Anti-circularity ledger for derivation v4.

2 Attempt A: Pressure-Balance Closure

Attempt 1 (Pressure-Balance Route). Use Eq. (2) to express ρ_P in terms of σ and R_ξ , then interpret ρ_P as an effective 5D cosmological constant Λ_5 to obtain κ_5^2 from the bulk Einstein equations.

2.1 Step A1: Express ρ_P from Pressure Balance

From Eq. (2):

$$\rho_P = \frac{\sigma}{2\pi R_\xi^2} \quad (3)$$

Dimensional check:

$$[\rho_P] = \frac{[\sigma]}{[R_\xi]^2} = \frac{L^{-4}}{L^2} = L^{-6} \quad (\text{WRONG!})$$

Problem: The dimensions do not match. We need $[\rho_P] = L^{-4}$, but $[\sigma/R_\xi^2] = L^{-6}$.

2.2 Step A2: Corrected Dimensional Form

The pressure-balance relation must include a thickness scale δ :

$$\sigma \sim \rho_P \cdot \delta \quad (4)$$

With the EDC postulate $\delta \sim R_\xi$:

$$\sigma = \alpha \cdot \rho_P \cdot R_\xi \quad (5)$$

where α is a dimensionless constant (the “ 2π ” in the original EDC formula likely absorbed other factors).

Dimensional check:

$$[\sigma] = [\rho_P][R_\xi] = L^{-4} \cdot L = L^{-3} \quad (\text{STILL WRONG!})$$

2.3 Step A3: Fundamental Dimensional Incompatibility

The issue is that $[\sigma] = L^{-4}$ (energy per 4-volume on the brane) while $[\rho_P \cdot \delta] = L^{-4} \cdot L = L^{-3}$ (energy per 3-volume).

For dimensional consistency, we need:

$$\sigma = \rho_P \cdot \delta^{-1} \cdot (4\text{D factor}) \quad (6)$$

or the original relation Eq. (2) must be reinterpreted as:

$$\sigma = 2\pi R_\xi^2 \cdot \rho_P \quad \Rightarrow \quad [\sigma] = L^2 \cdot L^{-4} = L^{-2} \quad (7)$$

which is inconsistent with $[\sigma] = L^{-4}$.

Resolution [P]: The EDC relation $\sigma = 2\pi R_\xi^2 \rho_P$ must use different unit conventions (e.g., σ as surface tension [L^{-2}] rather than 4D brane tension [L^{-4}]). This requires clarification from EDC source documents.

2.4 Step A4: Interpret ρ_P as Λ_5

Assuming we can express ρ_P in terms of σ and R_ξ with correct dimensions, we would identify:

$$\Lambda_5 = 8\pi G_5 \rho_P = \kappa_5^2 \rho_P \quad (8)$$

where we used $\kappa_5^2 = 8\pi G_5$.

However, this gives:

$$\kappa_5^2 = \frac{\Lambda_5}{\rho_P} \quad (9)$$

which is circular: we need Λ_5 to find κ_5^2 , but Λ_5 is defined via κ_5^2 .

Proposition 1 (Attempt A Failure). *The pressure-balance relation cannot fix κ_5^2 because:*

1. *The dimensional form requires clarification of EDC conventions.*
2. *Even with correct dimensions, linking ρ_P to Λ_5 introduces circularity ($\Lambda_5 \sim \kappa_5^2 \rho_P$).*
3. *No new dimensionless ratio emerges to fix C .*

Status: [OPEN] — Attempt A fails to fix C .

3 Attempt B: Induced Gravity with EDC UV Cutoff

Attempt 2 (Induced Gravity / Sakharov-Style). If EDC has no fundamental Einstein-Hilbert term, gravity could emerge as an induced effect from quantum fluctuations of matter fields. The induced gravitational coupling scales as $\kappa_5^{-2} \sim N\Lambda_{\text{UV}}^3$ in 5D, where N counts field species and Λ_{UV} is the UV cutoff.

3.1 Step B1: Induced Gravity in 5D

In the Sakharov/Adler approach, integrating out matter fields with UV cutoff Λ_{UV} generates an effective Einstein-Hilbert term:

$$S_{\text{induced}} = \frac{1}{2\kappa_5^2} \int d^5X \sqrt{-g^{(5)}} R^{(5)} \quad (10)$$

with the induced coupling:

$$\kappa_5^{-2} \sim \frac{N}{(4\pi)^{5/2}} \Lambda_{\text{UV}}^3 \quad (11)$$

where the numerical factor depends on regularization scheme.

Dimensional check:

$$[\kappa_5^{-2}] = [\Lambda_{\text{UV}}]^3 = L^{-3} \quad \checkmark$$

3.2 Step B2: Map Λ_{UV} to EDC Scale

EDC provides two candidate UV cutoffs:

Option B2a: $\Lambda_{\text{UV}} = R_\xi^{-1}$ (weak scale)

$$\kappa_5^{-2} \sim N R_\xi^{-3} \quad \Rightarrow \quad \kappa_5^2 \sim \frac{R_\xi^3}{N} \quad (12)$$

Option B2b: $\Lambda_{\text{UV}} = \sigma^{1/4}$ (tension-derived scale)

$$\kappa_5^{-2} \sim N \sigma^{3/4} \quad \Rightarrow \quad \kappa_5^2 \sim \frac{1}{N \sigma^{3/4}} \quad (13)$$

Option B2b matches the form $\kappa_5^2 = C \sigma^{-3/4}$ with $C = 1/N$.

3.3 Step B3: Can N Be Fixed?

To fix $C = 1/N$, we need EDC to specify the number of field species N .

Problem: EDC does not provide a first-principles count of N . Possible interpretations:

- $N = 1$ (single Plenum field) $\Rightarrow C \sim 1$
- $N \sim 10^2$ (Standard Model species) $\Rightarrow C \sim 10^{-2}$
- $N \sim ?$ (unknown EDC spectrum)

3.4 Step B4: Numerical Factors

Even fixing $N = 1$, the induced gravity formula contains scheme-dependent numerical factors:

$$\kappa_5^{-2} = \frac{N}{(4\pi)^{5/2}} \cdot f(\text{regularization}) \cdot \Lambda_{\text{UV}}^3 \quad (14)$$

where f depends on cutoff procedure (hard cutoff, dimensional regularization, etc.).

Proposition 2 (Attempt B Partial Success). *Induced gravity with $\Lambda_{UV} = \sigma^{1/4}$ reproduces the form $\kappa_5^2 = C\sigma^{-3/4}$ and identifies:*

$$C = \frac{(4\pi)^{5/2}}{N \cdot f} \quad (15)$$

However, C remains undetermined because:

1. N (species count) is not fixed by EDC.
2. f (regularization factor) is scheme-dependent.
3. EDC does not specify whether fundamental EH term exists (if it does, induced contribution is subleading).

Status: [OPEN] — Attempt B fixes C only up to $\mathcal{O}(1)$ factors.

4 Attempt C: Junction Curvature Normalization

Attempt 3 (Junction Curvature Route). From the Israel junction conditions, the characteristic extrinsic curvature scale is $k \sim \kappa_5^2 \sigma$. If EDC geometry specifies k in terms of intrinsic scales (e.g., $k \sim R_\xi^{-1}$), we can solve for κ_5^2 .

4.1 Step C1: Junction-Derived Curvature

From derivation v3, the Israel conditions with $\mathbb{Z}_2$ symmetry give:

$$K_{\mu\nu} = -\frac{\kappa_5^2}{2} \left(S_{\mu\nu} - \frac{1}{3} h_{\mu\nu} S \right) \quad (16)$$

For a pure-tension brane ($S_{\mu\nu} = -\sigma h_{\mu\nu}$):

$$K_{\mu\nu} = \frac{\kappa_5^2 \sigma}{6} h_{\mu\nu} \equiv k h_{\mu\nu} \quad (17)$$

Thus the curvature scale is:

$$k = \frac{\kappa_5^2 \sigma}{6} \quad (18)$$

Using $\kappa_5^2 = C\sigma^{-3/4}$:

$$k = \frac{C\sigma^{1/4}}{6} = \frac{C}{6\ell_\sigma} \quad (19)$$

where $\ell_\sigma = \sigma^{-1/4}$ is the tension-derived length scale.

4.2 Step C2: Attempt to Fix k from EDC Geometry

Hypothesis [P]: The extrinsic curvature scale k equals the intrinsic curvature of the EDC membrane:

$$k \stackrel{?}{=} \frac{1}{R_\xi} \quad (20)$$

Combining Eqs. (19) and (20):

$$\frac{C}{6\ell_\sigma} = \frac{1}{R_\xi} \quad \Rightarrow \quad C = \frac{6\ell_\sigma}{R_\xi} = 6 \cdot \frac{\sigma^{-1/4}}{R_\xi} \quad (21)$$

Dimensional check:

$$[C] = \frac{[\sigma^{-1/4}]}{[R_\xi]} = \frac{L}{L} = 1 \quad \checkmark$$

4.3 Step C3: Is Eq. (21) a Closure?

Eq. (21) expresses C in terms of σ and R_ξ :

$$\boxed{C = 6 \cdot \sigma^{-1/4} R_\xi^{-1}} \quad (22)$$

This would give:

$$\kappa_5^2 = C \sigma^{-3/4} = 6 \sigma^{-1/4} R_\xi^{-1} \cdot \sigma^{-3/4} = \frac{6}{\sigma R_\xi} \quad (23)$$

Dimensional check:

$$[\kappa_5^2] = \frac{1}{[\sigma][R_\xi]} = \frac{1}{L^{-4} \cdot L} = L^3 \quad \checkmark$$

4.4 Step C4: Why This Is Not a True Closure

The “closure” in Eq. (23) is **illusory** because:

1. **Hypothesis (20) is not derived.** The identification $k = 1/R_\xi$ is a *postulate*, not a consequence of EDC equations of motion.
2. **Circular reasoning.** If we *define* $k = 1/R_\xi$, we are effectively choosing C to make this true. This is equivalent to fitting κ_5^2 to produce a desired curvature scale.
3. **No dynamical justification.** EDC does not provide an equation that determines the embedding curvature from σ and R_ξ alone.

Proposition 3 (Attempt C Failure). *The junction curvature route produces a dimensionally consistent expression for C , but relies on the unproven postulate $k = 1/R_\xi$. Without a dynamical derivation of this relation, Attempt C does not constitute a true closure.*

Status: [OPEN] — Attempt C fails without additional EDC input.

5 No-Go Lemma

Lemma 1 (Dimensional Counting No-Go). *Let the available EDC inputs be $\{\sigma, \rho_P, R_\xi\}$ with dimensions:*

$$[\sigma] = L^{-4}, \quad [\rho_P] = L^{-4}, \quad [R_\xi] = L$$

subject to the constraint $\sigma = \alpha \cdot \rho_P \cdot R_\xi^n$ for some n and dimensionless α .

Then the number of independent dimensionful parameters is at most 2, while determining κ_5^2 and C independently requires 3 scales.

Conclusion: σ , ρ_P , and R_ξ cannot uniquely fix C without an additional normalization principle.

Proof. The pressure-balance relation ties σ , ρ_P , and R_ξ :

$$\sigma \sim \rho_P \cdot R_\xi^n$$

For dimensional consistency with $[\sigma] = [\rho_P] = L^{-4}$, we need $n = 0$, implying $\sigma \propto \rho_P$ (no independent information from R_ξ).

Alternatively, if $[\sigma] \neq [\rho_P]$, then $n \neq 0$ and we have:

$$\sigma = f(\rho_P, R_\xi)$$

reducing three parameters to two independent ones.

To construct the dimensionless $C = \kappa_5^2 \sigma^{3/4}$, we need:

$$C = g \left(\frac{\sigma}{\rho_P}, \frac{\sigma^{1/4}}{R_\xi^{-1}}, \dots \right)$$

But all such ratios are fixed by the pressure-balance constraint, leaving C undetermined up to a constant of order unity. $\square$

6 Summary and Outcome

6.1 Results by Attempt

Attempt	Method	Result	Status
A	Pressure-balance $\rightarrow \Lambda_5$	Dimensional issues + circularity	FAIL
B	Induced gravity	$C \sim 1/N$ but N unfixed	PARTIAL
C	Junction curvature	$C = 6\sigma^{-1/4}/R_\xi$ but unproven postulate	FAIL

Table 2: Summary of Attempt outcomes.

6.2 One-Line Outcome

NO-GO: The EDC parameters σ , ρ_P , and R_ξ provide only two independent scales. The dimensionless constant C in $\kappa_5^2 = C\sigma^{-3/4}$ cannot be fixed without an additional normalization principle or dynamical equation.

6.3 What Would Fix C ?

To close BLOCK-003, one of the following is needed:

1. **Additional dimensionless input:** A pure number from EDC topology (e.g., winding number, Euler characteristic, species count N).
2. **Dynamical equation for embedding curvature:** An EDC equation of motion that determines k from σ and bulk geometry.
3. **Normalization convention:** A canonical definition of σ that fixes the proportionality constant (e.g., σ defined such that $\ell_\sigma = R_\xi$).
4. **Alternative route:** Abandon $\kappa_5^2 = C\sigma^{-3/4}$ and derive κ_5^2 from a different principle (e.g., holographic, entropic).

6.4 BLOCK-003 Status

Remains OPEN.

The missing element has been narrowed from “derive κ_5^2 ” to “provide one additional normalization principle.”

Acknowledgments

This derivation attempt follows the BLOCK-003 program established in `paper_gravity_block003/main.tex` and builds on results from derivation v1 ($G_N = \kappa_5^2/(6\pi L)$), v2 ($L = R_\xi$), and v3 ($\kappa_5^2 = C\sigma^{-3/4}$, C undetermined).